

# SIYONA AGARWAL

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## EDUCATION

**Banasthali Vidyapith**, Jaipur, Rajasthan

**2022 - Present**

B.Tech in Computer Science - Artificial Intelligence | **CGPA: 9.35**

## WORK EXPERIENCE

**Thought2Designs System Pvt. Ltd. @ IIT Delhi**

**Jan 2026 - Present**

*Research and Development Intern*

**MMFT Simulator - Microfluidics IDE** | Python, C++, JS, Three.js

- Extended the TU Munich MMFT microfluidics engine with stochastic pump fluctuations, time-program simulations, and mixture genealogy tracing for advanced flow and concentration analysis.
- Built a browser-based microfluidic simulation IDE using Flask, JavaScript, and Three.js with OpenFOAM integration, 3D visualization, Monte Carlo tolerance analysis, and real-time simulation playback.
- Developed batch analysis, ParaView rendering, STL/ISO export pipelines, and cross-platform tooling for interactive microfluidic chip design and simulation workflows.

**Predictive Maintenance Digital Twin** | Python, PyTorch, Flask [Indian Army]

- Built a predictive maintenance pipeline for industrial bearing failure detection using LSTM, CNN, Transformer, and Autoencoder models on 648-hour sensor data.
- Engineered physics-informed features from vibration and thermal signals for fault classification and Remaining Useful Life (RUL) prediction.
- Developed a Flask-based inference dashboard for real-time failure prediction, anomaly visualization, and cross-domain validation on CWRU bearing datasets.

**MSIM - Multi-Paradigm Simulation IDE** | TypeScript, React, Python

- Built a cross-platform multi-paradigm simulation IDE using Electron, React, and FastAPI with visual drag-and-drop modeling for agent-based, system-dynamics, and hybrid simulations.
- Developed a real-time code generation and simulation pipeline with WebSocket-based live animation, multi-tab statechart editor, undo/redo system, and standalone Windows/Linux/macOS packaging with zero Python installation required.

## Growth Natives

**May - July 2025**

*AI/ML Summer Intern*

- Executed EDA, ML pipelines (Media Mix Modeling, PyCaret), and chatbot spam prevention; certified in SQL and Excel.

**Email Guesser: Intelligent Email Discovery Tool** | Python, MongoDB, Playwright

- Validated 10k+ contact records via pattern logic and MX lookup; chunked processing, MongoDB integration, and crash-resumable workflows.

## PROJECTS

**Praavinya: Placement and Career Suite** | Python, HTML, CSS, JS, SQLAlchemy

**Aug 2024 - Apr 2025**

- Resume builder with multi-step student profile forms, PDF uploads, and structured data handling.
- Selenium-based web scraping for live job listings with filtering and keyword matching.
- Admin panel for user and document management; awarded **Best UI/UX Project** among 100+ entries by ACM Student Chapter.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, JavaScript, TypeScript, SQL

**Frontend:** HTML, CSS, React, Three.js, Electron

**Backend:** Flask, FastAPI, Node.js

**ML/AI:** PyTorch, scikit-learn, NumPy, Pandas, PyCaret

**Databases:** MongoDB, SQLAlchemy

**Core CS:** Data Structures & Algorithms, DBMS, Computer Networks

## CERTIFICATIONS

**WACIIS Pre-Conference Workshop (IHCI)**

**Aug 30-31, 2024**

Presented paper poster: J.A.R.V.I.S. (Just A Random Very Intelligent System)

**Advanced Diploma in Networking, Cisco Networking Academy**

**2022 – 2024**